# INTRODUCTION

Prompt engineering is the process of designing and structuring inputs so that an Artificial Intelligence system can produce useful and accurate responses. Different types of prompts guide the AI in different ways. Some prompts ask for direct answers, some organize information in tables, some begin with questions that guide reasoning, and others require the AI to complete missing information.

## 1. Straightforward Prompts

A Straightforward Prompt is a simple and direct instruction given to an AI system. It asks for a clear and concise response without requiring complex reasoning or multiple steps. These prompts are commonly used when the user wants a quick definition, explanation, or short list.

Straightforward prompts are useful because they save time and provide focused answers. Since the instruction is direct, the AI can easily understand what information is needed and provide a precise response.

For example, the prompt “Define photosynthesis in one sentence.” asks the AI to provide a short and clear definition. Photosynthesis is the process by which green plants use sunlight, water, and carbon dioxide to produce food and release oxygen. This prompt type limits the answer to a single sentence, ensuring that the response is brief and easy to understand.

Another example is “List three advantages of electric vehicles.” This prompt instructs the AI to provide a list instead of a paragraph. Electric vehicles offer several benefits such as reduced air pollution, lower fuel costs, and quieter operation compared to traditional gasoline vehicles.

Straightforward prompts are widely used in education, research, and everyday applications. They help users quickly obtain essential information without unnecessary details. Because of their simplicity, they are one of the most common prompt types used in prompt engineering.

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## 2. Tabular Format Prompting

Tabular Format Prompting is a prompt type that asks the AI to present information in a table. Tables organize information into rows and columns, making it easier to compare multiple items.

This type of prompt is very useful when the user wants structured information or comparisons. Instead of writing long paragraphs, the AI arranges the data neatly so that readers can quickly understand the differences between items.

For instance, the prompt “Compare and contrast AC and DC current in a table.” asks the AI to organize the characteristics of Alternating Current and Direct Current in a structured format.

| Feature | AC Current | DC Current |
|---|---|---|
| Full Form | Alternating Current | Direct Current |
| Direction of Flow | Changes direction periodically | Flows in one direction |
| Source | Power stations | Batteries |
| Usage | Household electricity supply | Electronic devices |

This table clearly shows the differences between AC and DC current.

Another example is “Provide a table listing five programming languages, their paradigms, and one use case each.”

| Programming Language | Paradigm | Use Case |
|---|---|---|
| Python | Object-Oriented / Procedural | Data science and AI |
| Java | Object-Oriented | Enterprise applications |
| C | Procedural | System programming |
| JavaScript | Event-driven | Web development |
| SQL | Declarative | Database management |

Tabular prompting improves clarity and readability. It is especially helpful when dealing with technical comparisons, datasets, or structured information.

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## 3. Preceding Question Prompting

Preceding Question Prompting is a method where the prompt begins with a question that introduces a topic. The question encourages explanation or reasoning before presenting the final answer.

This technique helps guide the AI to explain a concept step by step. It also encourages deeper understanding rather than just providing a short definition.

For example, the prompt “Why is climate change a global concern? Explain how greenhouse gases contribute to global warming.”

Climate change is considered a global concern because it affects weather patterns, sea levels, ecosystems, and human life across the entire planet. Rising temperatures can lead to extreme weather events such as floods, droughts, and heat waves.

Greenhouse gases such as carbon dioxide and methane trap heat in the Earth's atmosphere. This process is known as the greenhouse effect. Normally, this effect helps keep the Earth warm enough to support life. However, excessive emissions from human activities like burning fossil fuels and deforestation increase the concentration of greenhouse gases, causing the Earth's temperature to rise.

Another example is "How do vaccines work? Describe the process of immunization in simple terms."

Vaccines work by preparing the immune system to recognize and fight harmful pathogens. A vaccine contains a weakened or inactive form of a virus or bacteria. When the vaccine enters the body, the immune system produces antibodies to fight it.

If the person is exposed to the real disease later, the immune system remembers the pathogen and quickly destroys it. This process is called immunization and helps protect individuals from serious infections.

Preceding question prompts are useful in educational settings because they encourage explanation and reasoning rather than short answers.

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## 4. Missing Word Prompting

Missing Word Prompting is a type of prompt where one or more words are intentionally left blank. The AI is expected to fill in the missing word based on context and knowledge.

This method is commonly used in quizzes, language learning, and knowledge testing. It checks whether the AI can understand the sentence and provide the correct missing information.

For example, the prompt "The capital of France is ____." requires the AI to fill in the blank with the correct answer, which is Paris.

Another example is "In photosynthesis, plants absorb sunlight to produce ____." The correct completion is glucose (food) and oxygen, as plants convert sunlight into chemical energy during photosynthesis.

Missing word prompts are effective for evaluating knowledge and reinforcing learning concepts. They are widely used in educational exercises and interactive learning systems.

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## Conclusion

Different types of prompts play an important role in guiding AI responses. Straightforward prompts provide direct answers, tabular prompts organize

information clearly, preceding question prompts encourage deeper explanations, and missing word prompts test knowledge by requiring completion of incomplete sentences.

Understanding these prompt types helps users interact with AI systems more effectively and obtain accurate, structured, and meaningful responses.